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The influence of artificial reef structural complexity on fish assemblage composition

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ABSTRACT

Purpose: built artificial reefs (ARs) are becoming an increasingly popular tool for enhancing recreational fishing activities. This study examined the establishment of fish assemblages at a new AR site in the Exmouth Gulf, Western Australia (King Reef). A remotely operated vehicle fitted with stereo-video (stereo-ROV) was used to survey the fish assemblages associated with four different artificial reef modules (Fish Towers, Apollos, Abitats, and Pyramids). The observation of recreationally targeted species at the AR provides evidence of its potential to contribute to local recreational fisheries. There were differences in the fish assemblages associated with the different modules with the Fish Towers (large subsurface steel buoys) having the highest mean fish abundance and biomass. The design of the modules that are incorporated into an AR design have a considerable influence in shaping the assemblage and biomass of fishes at an AR site. Some differences between module types were not immediately obvious due to the close spacing of modules which created a relatively contiguous fish assemblage across the whole reef field. These results indicate that the characteristics of purpose-built structures that influence the associated fish assemblage can be manipulated to create a design that can fulfil a specific purpose, such as promoting the recruitment and attraction of targeted species to enhance recreational fishing opportunities.

1. Introduction

The aquatic environment (encompassing all the dynamic and complex ecosystem processes) is experiencing the cumulative impacts of climate variability and a complex array of anthropogenic pressures. For example, there is pressure on a range of exploited fish stocks (Pauly et al., 1998, 2000; Myers and Worm, 2003). This global anthropogenic pressure includes impacts from commercial, recreational (anglers), and artisanal fishers. Recreational fishing activities (in isolation) may also have considerable impacts on fish populations and their wider ecosystems (Coleman et al., 2004; Altieri et al., 2012; Young et al., 2014; Lewin et al., 2019; Hyder et al., 2020), particularly in coastal areas close to large population centres, or popular holiday areas (McPhee et al., 2002). Recreational fishing activities are of high importance globally, particularly across developed countries and those with economies in transition (Freire et al., 2020; Arlinghaus et al., 2021), as they provide a range of social, economic, and health benefits including employment, education, tourism, exercise, relaxation, and social cohesion (Kearney, 2002; Cisneros-Montemayor and Sumaila, 2010; FAO, 2012). These benefits may not accrue or will not be evident if fish stocks, and their associated aquatic environment, become degraded. As such, the implementation of appropriate Ecosystem-Based Fisheries Management strategies (EBFM) that seek to mitigate adverse impacts (real or perceived), and/or seek to enhance the structure and functioning of whole aquatic ecosystems associated with recreational fishing activities (rather than single species fishery management) are of strategic benefit.

One EBFM strategy that has been promoted and trialled is the installation of artificial reefs (ARs), which are defined as submerged structures deliberately placed on the seafloor to mimic the characteristics of natural reefs, such as the provision of various ecosystem services (Baine, 2001; Cresson et al., 2014). ARs have been deployed to create a greater number of safe and more accessible fishing opportunities for recreational fishers (Becker et al., 2018; Florisson et al., 2018), to shift fishing pressure away from natural reefs (Leeworthy et al., 2006; Leitão

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et al., 2008), increase the catch of locally targeted species (Keller et al., 2016), and prevent illegal trawling (Sánchez-Jerez and Ramos-Esplá, 2000; Fabi et al., 2011; Brandini, 2014). The assumption that ARs will enhance fishery resources is based on the concept that the additional habitat provided by the AR will help overcome bottlenecks related to recruitment, habitat availability, or primary production (Becker et al., 2018). For example, the physical structure provided by an AR can increase settlement opportunities for autotrophic individuals and increase local upwelling and nutrient supplies, both of which support primary production (Leitão, 2013; Layman and Allgeier, 2020; Blount et al., 2021). Similarly, ARs provide recruitment sites for heterotrophic individuals that may otherwise have been absent from their overall population (Rilov and Benayahu, 2000; Layman and Allgeier, 2020). Both processes contribute to increased secondary production including that at higher trophic levels (i.e., targeted species; Simon et al., 2011; Leitão, 2013). Additionally, the structure of the AR provides shelter from predation (Bohnsack, 1989), which increases survival and feeding efficiency, promoting production through survival and growth (Pondella et al., 2022). ARs have the potential for high levels of production (Smith et al., 2016), depending on scale (i.e., number and size of ARs), which can promote increased catch rates and user satisfaction in recreational fishers (Mills et al., 2017; Florisson et al., 2018; Blount et al., 2021). However, increases in the abundance and biomass of fish may also be associated with attraction from nearby habitats, making targeted species easier to extract and more susceptible to overexploitation (Smith et al., 2015; Layman et al., 2016). ARs have been increasingly deployed for the purpose of enhancing recreational fishing, with large numbers of ARs present in the USA, and increasing numbers being installed in Australia (Ramm et al., 2021).

Until the 1980s, ARs were largely constructed from opportunistic materials, including rubble, rock, used tyres, old vehicles, and vessels and other steel structures (Sherman et al., 2002; Ramm et al., 2021), to simultaneously create large scale ARs and reuse waste materials. However, a greater understanding of the potentially harmful impacts of waste materials and the importance of AR design to their performance has resulted in a significant shift towards the use of purpose-built structures (Florisson et al., 2018; Ramm et al., 2021). Purpose-built structures are most commonly constructed from concrete (Baine, 2001), as it more closely mimics natural rocky substrate, can be moulded into various designs, and has fewer negative environmental impacts compared to other materials (Lima et al., 2019; Vivier et al., 2021). Concrete artificial reefs provide a substrate on which organisms can readily settle (Lima et al., 2019), and overall have been correlated with a high rate of efficacy and effectiveness (Vivier et al., 2021). Increasing numbers of steel structures from decommissioned oil and gas infrastructure are being intentionally repurposed as artificial reefs (often referred to as "rigs-to-reefs") (Bull and Love, 2019; Kaiser et al., 2020; Marnane et al., 2022; Sibley et al., 2023). These decommissioned structures have been shown to host large and dense fish assemblages and provide significant fishing opportunities (Boswell et al., 2010; Ajemian et al., 2015; Bollinger and Kline, 2017; Sibley et al., 2023).

The use of purpose-built structures allows the characteristics of the AR to be manipulated in order to create a design that can fulfil its intended purpose, which in a recreational fishing context, is increased fishing experiences and opportunities, and often an increased diversity and/or increased abundance of targeted species (Florisson et al., 2018). The structural characteristics, size, and location of an AR all shape its faunal assemblage as they influence feeding, shelter, and recruitment opportunities (Bohnsack et al., 1994; Rilov and Benayahu, 2000; Hackradt et al., 2011; Cresson et al., 2014; Davis and Smith, 2017; Holland et al., 2021). In particular, the structural complexity of an AR, encompassing features like holes and crevices, vertical relief and rugosity, is influential in shaping its fish assemblage (Rilov and Benayahu, 2000; Charbonnel et al., 2002; Sherman et al., 2002; Hackradt et al., 2011; Paxton et al., 2017; Garner et al., 2019; Lemoine et al., 2019; Cardoso et al., 2020). However, these effects are generally species-specific, with different species preferring different design features (Komyakova et al., 2019). Consequently, different module designs may be expected to support different fish assemblages, and in turn, different recreationally targeted species. It is important to assess how the fish assemblages and populations of targeted fishes differ with AR designs, enabling decisions about how distinctive designs and features may be used to meet specific goals, such as recreational fisheries enhancement.

King Reef is an AR which was deployed in the Exmouth Gulf, Western Australia in July 2018, with the goal of enhancing local recreational fishing opportunities, primarily by providing a safe, accessible, and enjoyable fishing experience. The location and design of King Reef was intended to provide a mosaic of habitats to support recreationally targeted fish species in a safe accessible location to increase user satisfaction (Florisson et al., 2020). The reef consists of four different module types, each with a different size and shape that provide varying amounts of structural complexity. The objective of this research was to determine how the characteristics of four different reef modules influenced the fish assemblage associated with each module, the numbers of species, individual fish and overall biomass of fish. Also, the numbers of targeted fishes. We hypothesised that the larger and more structurally complex module designs comprising King Reef will support a greater number, diversity, and biomass of fish species, including those that are recreationally targeted.

2. Materials and methods

2.1. Study area

King Reef is located in Exmouth Gulf in the Gascoyne region of northwestern Australia, approximately five km offshore, and 6.5 km from the town of Exmouth (Fig. 1). The Gulf is a shallow (maximum depth of 22 m) tropical embayment and hosts a highly dynamic environment characterised by strong seasonal sea surface temperature (SST) variations, tidal-influenced hydrodynamics (Orpin et al., 1999), seasonal tropical cyclones, low primary productivity and fluctuating turbidity (Fromont et al., 2016; Cartwright et al., 2021). The Gulf supports a range of tropical and subtropical fish species, although detailed information about its fish assemblages are scarce (McLean et al., 2016). The benthic habitat in the Gulf consists largely of soft sandy sediment in addition to areas of limestone pavement, macroalgae, mangroves, seagrass, with limited high-relief reef on several shoals (Lyne et al., 2006). The gulf and its various marine habitats have been subject to a range of anthropogenic pressures resulting from activities including commercial (prawn trawling) and recreational fishing, tourism, shipping, and mining (Sutton and Shaw, 2021).

Surveys were conducted at King Reef in November 2018 and September 2019. The AR sits in approximately 18 m of water, covers approximately 8100 m^2 and consists of six Fish Towers (repurposed steel mid-water depth buoys) surrounded by 49 smaller concrete purpose-built modules of three distinctive designs (Fig. 2). The Apollos (n = 40; Fig. 2A) are 1 m high hollow cone shaped modules deployed in groups of four (i.e., a total of 10 clusters). The Abitats (n = 3; Fig. 2B) are low triangular prism-shaped modules, 1 m in height. The Pyramid modules (n = 6; Fig. 2C) are 2.1 m high with an open pyramid design. The Fish Tower structures (n = 6; Fig. 2D) are approximately 6 m high and consist of a cylinder approximately 3.7 m in diameter and 9–10 m long, with four attached arches on either side, and some additional complex structure on the top. The cylinder is raised above the seabed to create a cave-like space. Distances between neighbouring modules range between approximately three and 48 m.

2.2. Sampling options

Many artificial reefs are small in their physical size and spatial extent (Pondella et al., 2022) making it challenging to develop a robust

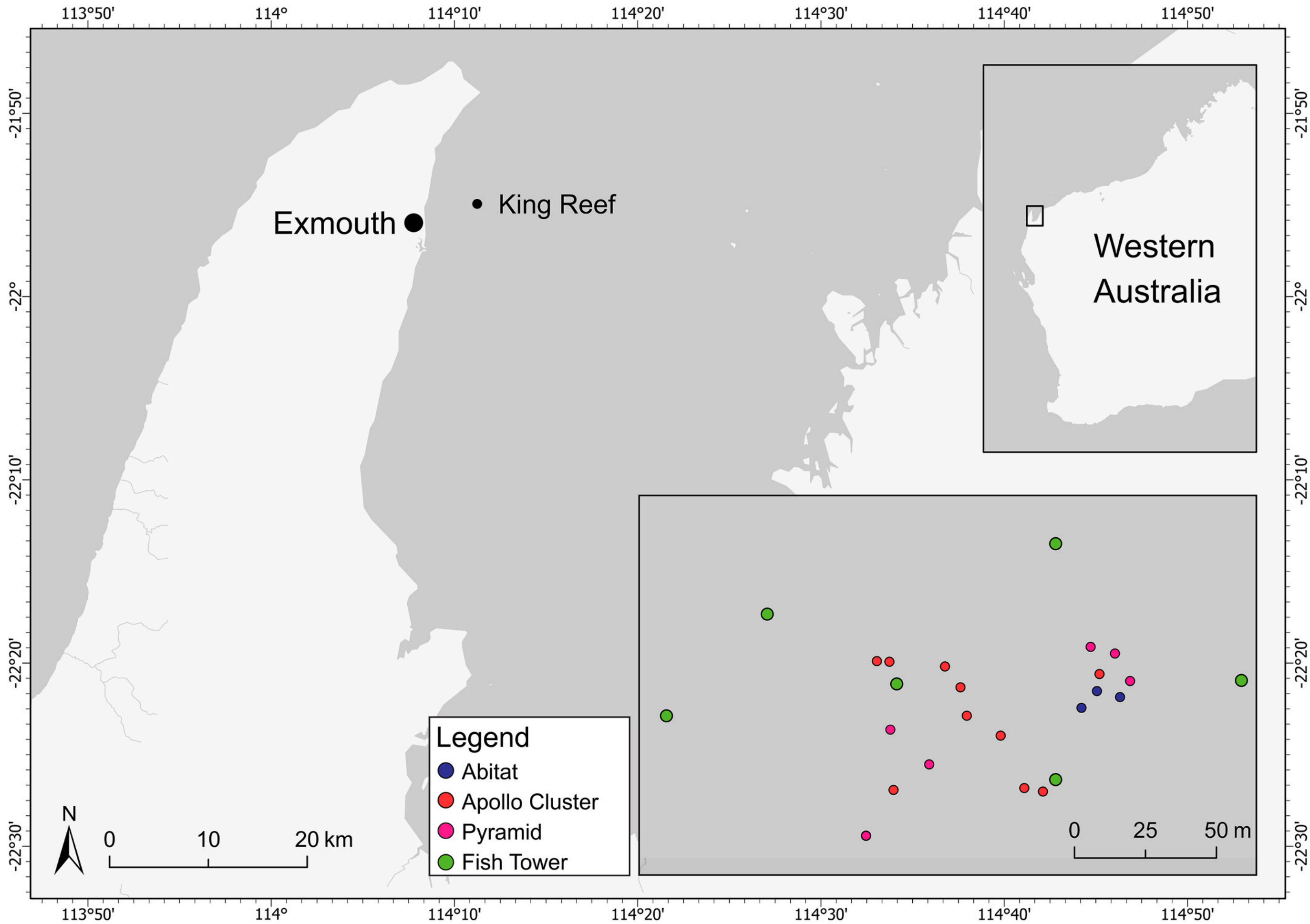

Fig. 1. Location of King Reef and its constituent modules within Exmouth gulf, Western Australia.

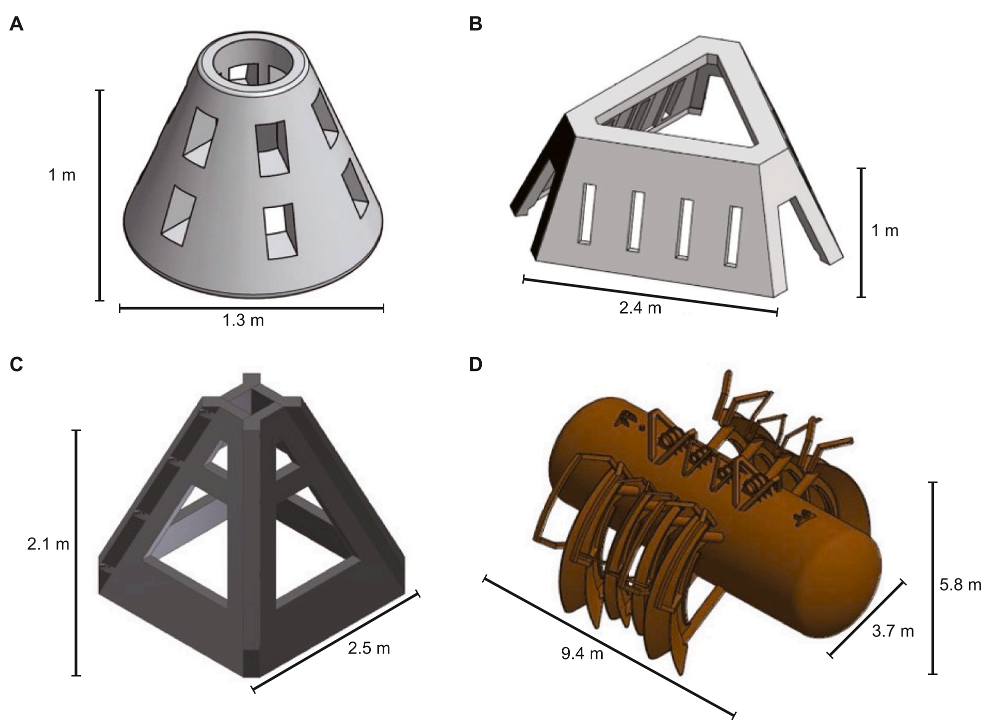

Fig. 2. Design of the four artificial reef modules comprising King Reef: Apollos, set in clusters of four (A), Abitat (B), Pyramid (C), and Fish Tower (D).

experimental design that samples only the fish associated with particular structures. We chose not to use any form of destructive sampling due to the possibility of repeated sampling causing depletion. We did not consider the use of visual transect-based methods as even short transects (e.g., 20 m long) would have encountered different module types. While baited remote video systems have become a popular sampling tool, the recommendation of having 500 m spacing between replicate stations to minimise overlap in the bait plume and potential pseudo-replication (Cappo et al., 2001) means we would not have been unable to investigate differences between module types (see Fig. 1). To obtain sufficient replicates to undertake a quantitative statistical analysis, a point count sampling approach was used with sampling undertaken by an ROV equipped with a stereo-video system. This sampling approach has been shown to be effective on other manmade structures which are limited in size (Harvey et al., 2021b).

2.2.1. Stereo-ROV

A small ROV (Blue ROV2 Heavy Configuration; 457 mm × 575 mm × 254 mm, ~11 kg) equipped with a stereo-video system was used to sample the reef fish assemblages. The stereo-video system consisted of two Sony Action cameras (FDR-X3000) mounted to the skid plate of the ROV, 590 mm apart and at a 5° inward angle (Harvey et al., 2010). The cameras recorded at a resolution of 1920 × 1080 p at 60 fps to account for vehicle movement. A SEATRAC Ultra Short Baseline positioning system (USBL; Seatrac X150 USBL and X010 Modem) was used for navigation and recording the position of the ROV. The USBL system had a range resolution of ±0.1, and angular resolution of ~2 % of the acoustic range (~1 m in this case).

2.2.2. Artificial reef surveys

Structures were sampled via 10 s snapshots of footage, during which the ROV paused and filmed a specific section of the structure for 10 s. Snapshots were taken at: a) the first encountered side of Pyramids and Abitats; b) the first encountered side of an Apollo cluster, followed by the opposite side; and c) at least three sides of the cylindrical shape of the Fish Tower (both long sides and one end) at both the bottom and top of the structure. Given the relatively small footprint of the Abitats, Apollos, and Pyramids, small numbers of each structure type, and close proximity of all reef components, this sampling method was chosen to limit pseudo-replication and to increase statistical power through increased replication. This sampling method, and the different numbers of each structure type in the reef meant that the number of replicates differed between structures. However, at least three replicate samples from each structure type were obtained in each year (Table 1). Sampling was performed between 7:30 and 16:00 h to minimise diurnal variations in fish assemblages (Myers et al., 2016).

2.2.3. Stereo calibration, video analysis and biomass conversion

Calibration of each stereo-video system was performed before and after sampling using the software 'CAL' (https://www.seagis.com.au/bundle.html), and following the methods outlined by Harvey and Shortis (1998). Footage was analysed using the EventMeasure Stereo software (https://www.seagis.com.au/event.html), during which all fish within each 10 s snapshot were identified, counted, and measured where possible given the constraints of visibility and fish movement. Fish counts and measurements were limited to individuals within 7 m of the cameras to ensure appropriate levels of accuracy and precision (Harvey et al., 2010), and to maintain a relatively consistent sample volume. Fish that were identified as having left and re-entered the area of the snapshot were only counted once. Where a species-level identification was not possible, identifications were pooled to the next possible taxonomic level (i.e., genus level) or grouped.

Length measurements were used to calculate fish biomass from documented length-weight relationships (Wilson et al., 2018). Where length-weight relationships were not available for a species, a relationship from a similar species was applied. Where an individual could not be measured (for example, due to partial obscuration or being front-on to the cameras), the mean length of the same species at the same structure type was applied to that individual.

Table 1
Number of replicate samples for each structure type and year.

Structure	November 2018	September 2019
Abitat	3	3
Apollo	15	18
Pyramid	7	6
Fish Tower	40	40

2.3. Statistical analysis

All statistical analyses were performed using Primer 7 with the PERMANOVA + add on (Anderson et al., 2008). Differences in fish assemblage composition between structures (4 levels, fixed: Abitat, Apollo, Fish Tower, and Pyramid) and years (2 levels, fixed: November 2018, and September 2019) were tested with a two-way PERMANOVA. The PERMANOVA was performed on a Bray-Curtis resemblance matrix constructed from zero-adjusted, dispersion weighted, and square root transformed data. These transformations were applied to allow the Bray-Curtis matrix calculation (Clarke and Gorley, 2015), to down weight the influence of highly abundant species over rare species (Clarke and Gorley, 2015), and to reduce the effects of numerically dominant, but patchy species (Clarke et al., 2006; Clarke and Gorley, 2015). The dispersion of the assemblage data was tested using PERMDISP (Anderson, 2006) after transformation, with no significant difference in time, structure, or time x structure being detected. Significant interactions and differences for the PERMANOVA were determined when $P < 0.05$ using 9999 permutations of the data. Where significant main effects or interactions occurred, pairwise comparisons were performed to determine the location of the differences. To further interpret the results of these formal tests, a canonical analysis of principal coordinates (CAP) ordination was constructed, overlaid with vectors of key species driving the ordination (Pearson's correlations to the CAP axes >0.3). Leave one out allocation (LOOA) tests were also completed to further investigate the degree of separation between fish assemblages at different structures.

Differences in fish abundance, the number of species, biomass, and the abundance of focal species between structures and years were tested with separate two-way PERMANOVAs using the same design described above. A priori, three focal species, *Lutjanus sebae* (Red emperor), *Gnathanodon speciosus* (Golden trevally), and *Lethrinus laticaudis* (Grass emperor) were chosen as they are commonly targeted by recreational fishers in the region (Ryan et al., 2019, 2022), and were observed in sufficient numbers to allow robust analysis. A SIMPER analysis was used to identify dominant species for further analysis. These were *Pentapodus porosus* (Northwestern threadfin bream), *Chelmon marginalis* (Margined coralfish), *Sardinella gibbosa* (Goldstripe sardinella), *Upeneus tragula* (Bartail goatfish), and *Lutjanus carponotatus* (Stripey snapper). Analyses were performed on Euclidean distance matrices. Where significant differences or interactions occurred in the main PERMANOVA test ($P < 0.05$), pairwise tests were performed to identify the location of the differences. Homogeneity of variance was tested prior to analysis using PERMDISP. Where significant differences in dispersion were detected, data were transformed with square root, 4th root or Log (x+1) transformations. Raw data for *U. tragula* met the assumptions of PERMDISP and analysis was undertaken on raw data. *Chelmon marginalis*, *L. sebae*, *P. porosus*, *G. speciosus, L. carponotatus,* and *L. laticaudis* did not meet the assumptions of dispersion for any transformation and analysis was undertaken on the raw data. Similarly, data for *S. gibbosa* did not meet the did not meet the assumptions of dispersion for any transformation and analysis was undertaken on square root data due to the high numbers of individuals in some samples.

3. Results

Across all structures and both years, a total of 2196 individuals from 63 unique taxa and 27 families were identified (Table 2), comprising 700 kg of biomass. The two most abundant species, accounting for 47 % of total fish abundance, were *Sardinella gibbosa* (n = 596), which were only observed at the Fish Towers, and *Sphyraena obtusata* (Obtuse barracuda; n = 432), which was recorded on the Fish Towers and Apollos. The greatest number of species (47) were observed at the Fish Towers, of which 22 were unique, followed by the Apollos (32 total, and 7 unique), Pyramids (23 total, and 3 unique), and Abitats (8 total, and 1 unique). We observed ten recreationally targeted species commonly caught in the region. These included *Lethrinus nebulosus* (Spangled emperor), *L. laticaudis*, *L. sebae*, *L. carponotatus*, *Symphorus nematophorus* (Chinamanfish), *G. speciosus*, *Plectropomus* spp. (Coral trout), *Epinephelus coioides* (Goldspotted rockcod), *Epinephelus rankini* (Rankin cod) and *Choerodon schoenleinii* (Blackspot tuskfish).

Table 2
Abundance of species observed at each of the structures comprising King Reef. Species marked with * are commonly targeted by recreational fishers in the region.

Family	Species	Common Name	Abitat	Apollo	Fish Tower	Pyramid
Carangidae	*Gnathanodon speciosus* *	Golden trevally	6	6		13
	Selaroides leptolepis	Yellowstripe scad		9	115	
	Seriola dumerili	Amberjack				1
	Seriola rivoliana	Highfin amberjack			12	
	Turrum fulvoguttatus	Turrum			1	
Chaetodontidae	*Chaetodon assarius*	Western butterflyfish			1	
	Chelmon marginalis	Margined coralfish	2	2	3	5
	Coradion chrysozonus	Orangebanded coralfish		1	6	1
	Heniochus acuminatus	Longfin bannerfish			5	
	Parachaetodon ocellatus	Ocellate butterflyfish			1	
Clupeidae	*Sardinella gibbosa*	Goldstripe sardinella			596	
Echeneidae	*Echeneis naucrates*	Sharksucker			2	
Ephippidae	*Platax batavianus*	Humphead batfish			1	
	Platax orbicularis	Round batfish			1	
Epinephelidae	*Epinephelus bilobatus*	Frostback rockcod		1		
	Epinephelus coioides *	Goldspotted rockcod			8	
	Epinephelus lanceolatus	Queensland grouper			1	
	Epinephelus rankini *	Rankin cod			2	
	Plectropomus spp *	Coral trout	1		2	
Glaucosomatidae	*Glaucosoma magnificum*	Threadfin pearl perch			155	1
Haemulidae	*Diagramma pictum labiosum*	Painted sweetlips			1	
	Plectorhinchus gibbosus	Brown sweetlips			2	
	Plectorhinchus multivittatus	Manyline sweetlips		2	1	
Holocentridae	*Sargocentron spp*	Squirrelfish				1
Labridae	*Choerodon cauteroma*	Bluespotted tuskfish		2	19	1
	Choerodon cephalotes	Purple tuskfish		14		1
	Choerodon schoenleinii *	Blackspot tuskfish		1	2	
	Choerodon vitta	Redline tuskfish			4	1
	Labroides dimidiatus	Bluestreak cleaner wrasse			2	
	Thalassoma lunare	Moon wrasse			1	
Latidae	*Psammoperca datnioides*	Black sand bass		22	135	3
Lethrinidae	*Lethrinus laticaudis* *	Grass emperor	2	19	36	3
	Lethrinus nebulosus *	Spangled emperor			4	
	Lethrinus punctulatus	Bluespotted emperor			1	
Lutjanidae	*Lutjanus carponotatus* *	Stripey snapper		8	21	2
	Lutjanus erythropterus	Crimson snapper			1	
	Lutjanus lemniscatus	Darktail snapper		1		
	Lutjanus malabaricus	Saddletail snapper			5	1
	Lutjanus russellii	Moses' snapper		2	11	
	Lutjanus sebae *	Red emperor		29	23	3
	Lutjanus vitta	Brownstripe snapper		6	2	1
	Symphorus nematophorus	Chinamanfish		1		
Monacanthidae	*Aluterus scriptus*	Scrawled leatherjacket		1		
	Monacanthus chinensis	Fanbelly leatherjacket		2		
	Paramonacanthus choirocephalus	Pigface leatherjacket		2	1	1
Mullidae	*Parupeneus chrysopleuron*	Rosy goatfish			1	
	Parupeneus indicus	Yellowspot goatfish		2	1	
	Parupeneus spilurus	Blackspot goatfish		1		
	Upeneus tragula	Bartail goatfish	4	14	62	29
Nemipteridae	*Pentapodus porosus*	Northwest threadfin bream	1	27		9
	Pentapodus vitta	Western butterfish		2		3
	Scaevius milii	Coral monocle bream		19	9	2
	Scolopsis monogramma	Rainbow monocle bream	1			
Pinguipedidae	*Parapercis nebulosa*	Pinkbanded grubfish		1		
Platycephalidae	*Platycephalus westraliae*	Yellowtail flathead		1	2	
Pomacanthidae	*Chaetodontoplus duboulayi*	Scribbled angelfish			6	1
	Pomacanthus sexstriatus	Sixband angelfish				1
Pomacentridae	*Abudefduf bengalensis*	Bengal sergeant		6	4	
Scorpaenidae	*Pterois volitans*	Common lionfish			3	
Siganidae	*Siganus* sp1	Rabbitfish		12	147	2
Sparidae	*Rhabdosargus sarba*	Tarwhine	1	1		
Sphyraenidae	*Sphyraena obtusata*	Obtuse barracuda		12	420	
Terapontidae	*Helotes octolineatus*	Western striped grunter			24	

3.1. Fish assemblage composition

Although no interaction was detected between structure and year (*Structure x Year*: $F_{3,\,131} = 1.345$, $P < 0.05$), fish assemblages did differ between structures (*Structure*: $F_{3,\,131} = 2.856$, $P < 0.001$), and between the two years (*Year*: $F_{1,\,131} = 3.210$, $P < 0.001$). Pairwise tests indicated that the assemblages differed between the Fish Towers and Apollos ($P_{(pairwise)} < 0.001$), Fish Towers and Pyramids ($P_{(pairwise)} < 0.01$), and the Apollos and Abitats ($P_{(pairwise)} < 0.05$), but were similar across all remaining pairwise combinations ($P_{(pairwise)} > 0.05$). These results were supported by the CAP ordination (Fig. 3), and LOOA tests (Table 3). The ordination showed some separation of the Fish Towers and Apollos, and this was reflected in their individual LOOA successes (80 % and 60.6 %, respectively). In contrast, the assemblages associated with the Pyramid and Abitat structures had a high amount of overlap with that of other structures, as indicated by their lower LOOA success (15.4 % and 50 %, respectively). Indeed, 70 % of mis-classified samples were incorrectly identified as belonging to either Abitats or Pyramids. The vectors overlaid on the CAP ordination (Fig. 3) indicate that fish assemblages at the Fish Towers were characterised by high abundances of *S. gibbosa* and *S. obtusata*, and a lack of *Pentapodus porosus* and *Choerodon cephalotes* (Purple tuskfish), which drove their separation from the other three structure types. The Apollos were further separated from the Abitats and Pyramids through their greater abundances of *Scaevius milii* (Coral monocle bream), *Abudefduf bengalensis* (Bengal sergeant) and *L. sebae*. Similarly, the separation of the Pyramids from the Apollos and Fish Towers was driven by *C. marginalis* and *U. tragula*.

3.2. Fish abundance, number of species, and biomass

The mean number of species per sample did not vary between structures (*Structure*: $F_{3,\,131} = 1.471$, $P = 0.223$; Fig. 4A) or years (*Year*: $F_{1,\,131} = 3.245$, $P = 0.072$). Mean fish abundance was similar across years (*Year*: $F_{1,\,131} = 1.899$, $P = 0.170$), but varied significantly between structures (*Structure*: $F_{3,\,131} = 5.782$, $P < 0.01$), with the Fish Towers having a greater number of fish than all other structures ($p_{(pairwise)} < 0.05$), which in turn were all similar ($P_{(pairwise)} > 0.1$; Fig. 4B). Conversely, the mean fish biomass per sample was significantly higher in 2018 than 2019 (*Year*: $F_{1,\,131} = 6.510$, $P = 0.028$), but did not differ between structures (*Structure*: $F_{3,\,131} = 1.744$, $P = 0.159$).

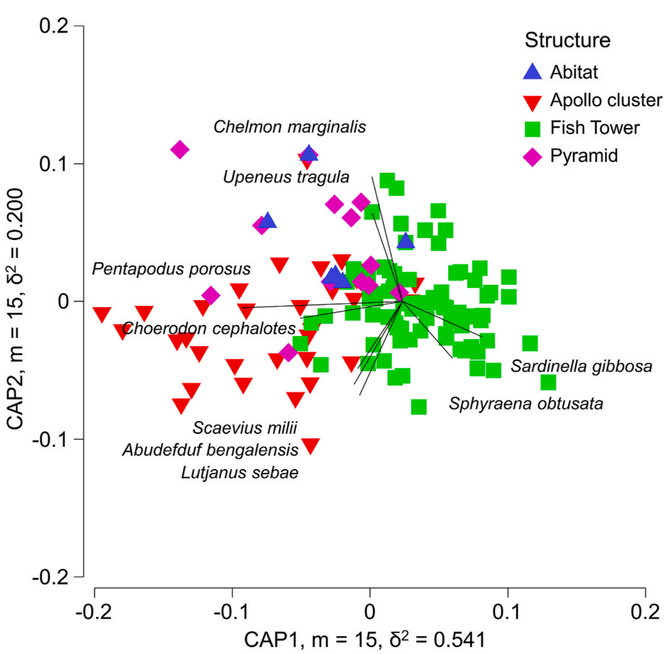

Fig. 3. Canonical Analysis of Principal Coordinates (CAP) ordination of fish assemblages observed across the four artificial structure types comprising King Reef. Species with Pearson's correlation >0.3 to the CAP axes are overlaid as vectors. Analysis based on dispersion weighting and square root transformation and Bray-Curtis similarities.

Table 3
Leave one out allocation success of fish assemblage observations to structure type. Values in bold indicate the number of correct classifications.

Original Structure	m = 15. Total correct: 89/132 (67.4 %)					
	Classified					
	Abitat	Apollo	Fish Tower	Pyramid	Total	% correct
Abitat	**3**	0	1	2	6	50
Apollo	7	**20**	3	3	33	60.61
Fish Tower	10	4	**64**	2	80	80
Pyramid	6	2	3	**2**	13	15.39

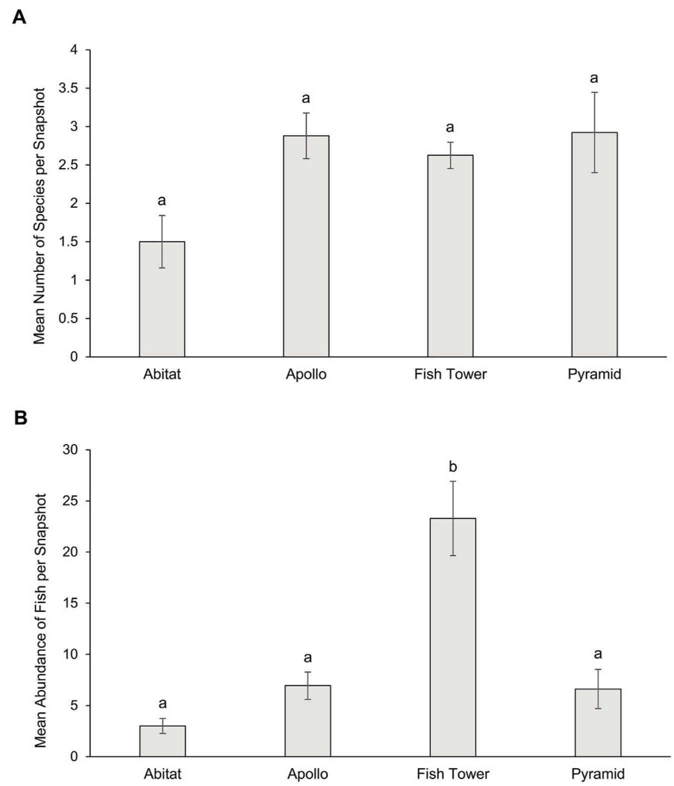

Fig. 4. Mean (±1SE) fish abundance (A), and species richness (B) across different artificial reef structure types: Abitat (n = 6), Apollo (n = 33), Fish Tower (n = 80) and Pyramid (n = 13). Statistically similar means are indicated by shared letters.

3.3. Focal species

Two of the focal species, *C. marginalis* and *P. porosus*, had significantly different levels of abundance between structure types ($F_{3,\,131} = 4.5788$, $P < 0.05$, and $F_{3,\,131} = 11.634$, $P < 0.001$, respectively). *Chelmon marginalis* was more abundant on the Abitat and Pyramid structures than the Fish Towers and Apollos ($P_{(pairwise)} < 0.05$; Fig. 5A), whilst *P. porosus* was more abundant on the Abitats, Apollos, and Pyramids than on the Fish Towers, where it was absent ($P_{(pairwise)} < 0.05$; Fig. 5B). The abundances of the remaining focal species were not significantly different between structures (Fig. 5C–H). However, there were some noteworthy trends. For example, none of the targeted focal species showed distinct preferences for any structure, although *L. sebae* and *L. carponotatus* were absent at the Abitats (Fig. 5F and G), and

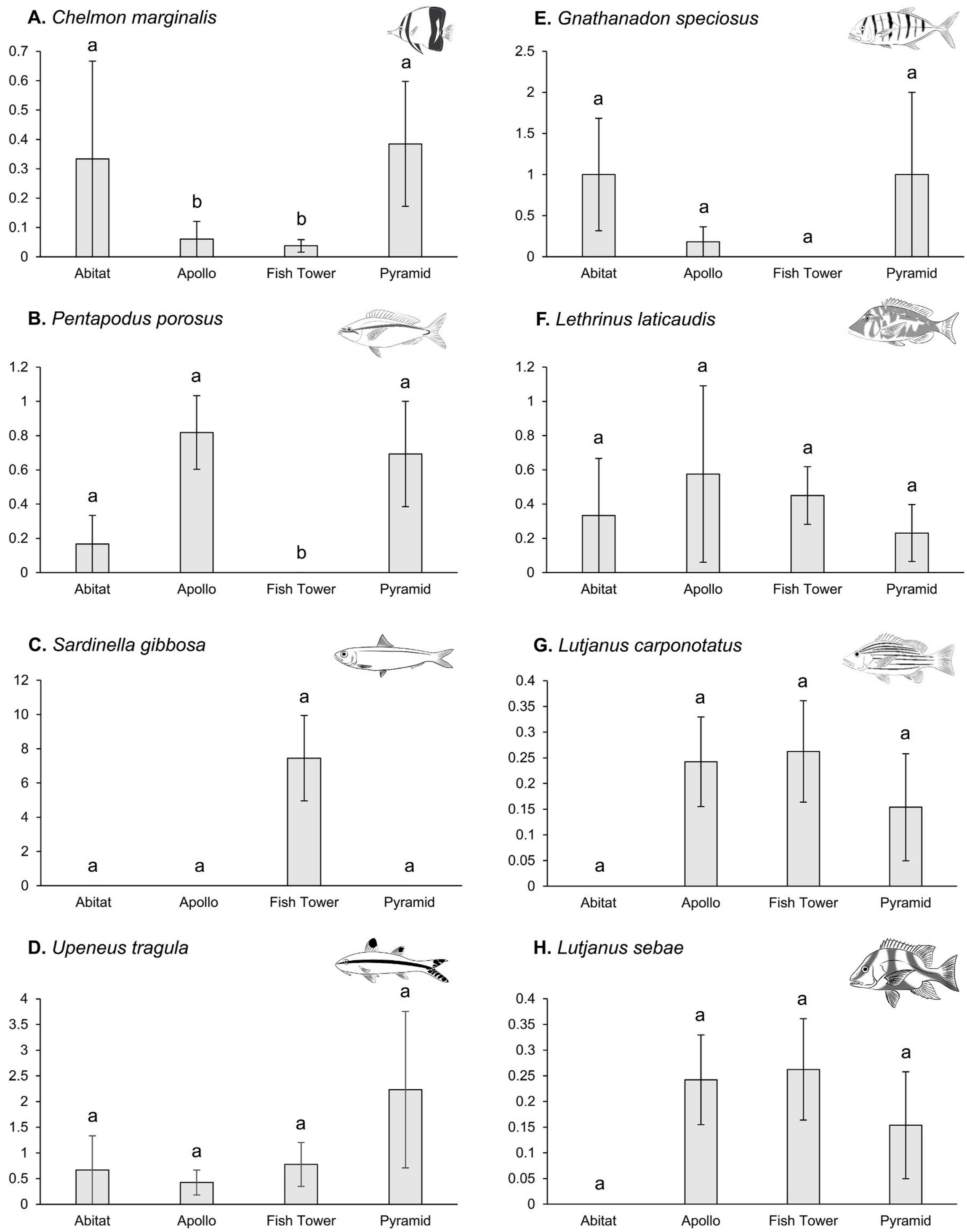

Fig. 5. Mean abundance (±1SE) of focal species: *Chelmon marginalis* (A), *Pentapodus porosus* (B), *Sardinella gibbosa* (C), *Upeneus tragula* (D), *Gnathanodon speciosus* (E), *Lethrinus laticaudis* (F), *Lutjanus carponotatus* (G), and *Lutjanus sebae* (H); across different artificial reef structure types: Abitat (n = 6), Apollo (n = 33), Fish Tower (n = 80) and Pyramid (n = 13). Statistically similar means are indicated by shared letters.

G. speciosus was not observed at the Fish Towers (Fig. 5E). Similarly, *S. gibbosa*, a schooling planktivore, was observed only at the Fish Towers (Fig. 5C).

4. Discussion

Artificial reefs are being increasingly deployed to enhance recreational fisheries, but it is important to ensure their designs are fit for purpose. Assessing the fish assemblages associated with different AR designs can provide insights into how designs can be manipulated to allow the outcomes of the AR to be optimised for a range of different stakeholders and purposes. For King Reef, the designs were chosen to support a diverse fish assemblage, including recreationally targeted species. Our comparison of the fish assemblages associated with the four types of structures showed there was variation in the composition of the fish assemblages associated with each structure. This was most probably caused by the differences in structural complexity between the designs. The fish assemblages associated with the Fish Towers and Apollos were different to each other and overall were the most unique, although neither was completely distinct from all other structures. The Abitats and Pyramids had intermediate assemblages, which were similar to each other and to the Fish Towers and Apollos, respectively. Notably, while the mean number of species did not differ statistically between structures, the Fish Towers had a total number of species 1.4 times greater than the structure with the next highest total number of species (Apollos) and had at least three times as many unique species as the other structures. Additionally, mean fish abundance was between 3.3 and 7.8 times higher at the Fish Towers compared to the other modules. Importantly, ten common recreationally targeted species in the region were observed at the reef.

The Fish Tower structures were notable not only because of their unique assemblage composition, but also because of their high fish abundance, total number of species, and number of unique species compared to the smaller purpose-built modules. The larger Fish Towers had a greater abundance and variety of predatory species from families including Epinephelidae, Lutjanidae, and Lethrinidae. The Fish Towers also had unique structural characteristics compared to the Apollos, Abitats, and Pyramids. Namely, these were greater volume, vertical relief, material composition (steel rather than concrete) and structural complexity. Design characteristics of ARs influence various aspects of their associated fish assemblages (Charbonnel et al., 2002; Sherman et al., 2002; Hackradt et al., 2011; Komyakova et al., 2019; Lemoine et al., 2019; Hylkema et al., 2020; Blount et al., 2021; Pondella et al., 2022), and the differences in these features may have shaped the differences in fish community metrics between structures.

Vertical relief is well-recognised as a feature of man-made structures that influences their fish assemblages (Blount et al., 2021; Pondella et al., 2022), and was likely a key factor in separating the Fish Towers from other modules (although we note this may be an artefact of the greater number of samples). ARs with high vertical relief can support greater densities of fish and create variation in densities of different trophic guilds, overall creating different assemblages to those associated with structures with less vertical relief (Pondella et al., 2022). For example, greater vertical relief can alter local hydrodynamics to increase the local supply of phytoplankton and zooplankton while providing habitat for zooplanktivorous fishes to feed and shelter, supporting their increased abundances (Champion et al., 2015; Blount et al., 2021; Vivier et al., 2021; Pondella et al., 2022). This was reflected in the greater abundances of the zooplanktivorous *Sardinella gibbosa* observed at the Fish Towers, which contrasted with the other low vertical relief structures where *S. gibbosa* was not observed. Schooling zooplanktivores are an important trophic link for piscivorous fishes, including recreationally targeted species (Pondella et al., 2022), several of which were observed at the Fish Towers. Structures with high vertical relief may attract and support higher abundances of pelagic predators, as they host larger amounts of prey and are easier to observe amongst bare sand (Paxton et al., 2020; Pondella et al., 2022). This was observed with *Turrum fulvoguttatus* and *Sphyraena obtusata*, which were observed at the Fish Towers either exclusively or in high abundances. Incorporating areas of high vertical relief into an AR design may be beneficial with greater vertical relief known to support a more diverse and abundant fish assemblage and a more complete food web. This in turn enhances recreational fishing experiences by supporting greater numbers of predatory fishes (which are often target species).

The higher total number of species and number of unique species associated with the Fish Towers may have been driven by its greater structural complexity (noting the higher level of replication). This concept was also supported by Cardoso et al. (2020), who found a greater diversity of both rare and typical species in high complexity areas of ARs. Similarly, other researchers have found positive relationships between structural complexity and fish abundance (Charbonnel et al., 2002; Sherman et al., 2002; Hunter and Sayer, 2009; Hackradt et al., 2011; Lemoine et al., 2019; Hylkema et al., 2020). Greater complexity increases the surface area available for the settlement of algae and sessile invertebrates, increasing the abundance of food resources, microhabitats, and therefore ecological niches (Blount et al., 2021; Pondella et al., 2022). Structures with higher numbers and a variety of different sized crevices, holes, and void spaces increase the shelter for different-sized species and life-stages, promoting increased species richness (Sherman et al., 2002; Pondella et al., 2022). Both of these pathways may have allowed the Fish Towers to support a greater variety of species, including those that are recreationally targeted. Unique species at the Fish Towers included various large sedentary predators, (e.g., cave dwelling *Epinephelus* spp.), which utilised the large void spaces and overhangs that were unique to the Fish Towers. When predators are present, structurally complex microhabitats provide shelter for prey species, allowing them to cohabit the area (Almany, 2004; Hensel et al., 2019), further contributing to increased species richness. Higher species richness can increase the diversity of catch for anglers (Blount et al., 2021). Therefore, incorporating complex designs into artificial reefs may be beneficial for recreational fisheries enhancement.

The fish assemblages at the Apollo, Pyramid, and Abitat modules were distinct from the Fish Towers, which may have been driven by their small size and low vertical relief (although this may also be due to their lower numbers of samples). Fish assemblages at these modules were characterised by a variety of sediment-associated species and mobile predators, in addition to several reef-associated species. This may reflect the different habitat and feeding resources provided by the smaller modules compared to the Fish Towers, on account of their size and complexity. The smaller size of the individual purpose-built modules gives them a higher perimeter to area ratio and more edge habitat (Bohnsack et al., 1994), which could explain the presence of both reef and sediment-associated species. The incidence of sediment-associated species (e.g., *Pentapodus porosus*, *Scaevius milii*, *Upeneus tragula*) increased around the purpose-built modules and was a key driver in separating their assemblages from that of the Fish Towers. Similarly, species that use both reef habitat and the habitat adjacent to reefs (e.g., *Choerodon* spp.; Fairclough, 2005) may prefer the small, low-relief modules where they can easily access both habitat types. For mobile predators, such as *Lutjanus carponotatus*, *Lutjanus sebae*, and *Gnathanodon speciosus*, small modules and the surrounding open sediment may be favourable foraging grounds as prey is more vulnerable in these habitats compared to the larger modules (Gatts et al., 2014), and foraging at the edges of these habitat types allows them to access resources from both habitats (Davenport et al., 2023). This highlights the potential importance of including smaller modules in the reef complex, as it may promote a variety of ecological niches that support a more speciose fish assemblage, including recreationally targeted species.

Evaluation of a species' preferences (or lack thereof) towards particular structure types is essential for determining how different structural features can be incorporated into ARs to promote the presence of recreationally targeted species. In the case of King Reef, this requires examining the relationships of targeted species with the different structures. Seven of the eight commonly caught targeted species observed were recorded at the Fish Towers, with the groupers (*Plectropomus* spp., *Epinephelus coioides* and *Epinephelus rankini*) and *Lethrinus nebulosus* observed almost exclusively at this structure (although this may be reflective of lower sampling effort at the other structures). For the larger-bodied, more sedentary groupers, the Fish Towers provide larger sized overhangs and caves that may form appropriately-sized refuges, which the smaller purpose-built modules lacked in comparison (Collins et al., 2015; Blount et al., 2021; Madgett et al., 2023). Among other targeted species from the Lutjanidae, Lethrinidae and Carangidae, however, different patterns emerged. *Lethrinus laticaudis* and *L. carponotatus* showed little preference for particular structures, and *G. speciosus* was found on the three smaller module types but not on the Fish Towers. The variation in structure preferences between different targeted species highlights the importance of including a variety of structural features across an AR so that a range of desirable recreational species can be accommodated.

The lack of difference between the fish assemblages of different structures may have been driven by the spatial distribution of the modules. Close spacing of modules can create a contiguous reef field encompassing all structures, with Becker et al. (2019) finding that 50 m spacing between clusters of small modules created a connected fish assemblage. Additionally, zones beyond which reef fish density decreases have been found at 30 m (Rosemond et al., 2018) and 50 m (dos Santos et al., 2010) away from ARs, meaning that an AR with structures separated by distances less than this would likely have consistent densities of reef associated species across their whole area. Close spacing between modules encourages wider fish movement as it provides more refuges across a greater foraging area, reducing the risk of predation when individuals move further across the reef field (Becker et al., 2019). All Abitat modules were placed next to Pyramid modules (with no more than 13.2 m separation), which may have contributed to the similarity between their fish assemblage compositions. Similarly, all Pyramid modules were located no more than 22 m from an Apollo structure. King Reef covers 8100 m^2, in which the Tower structures are spaced on average 70 m apart, with the smaller modules distributed amongst them. Importantly, the distances between different modules were small, such that the distance between adjacent modules did not exceed 50 m. The distance between individual modules would have been enough to create a singular reef field, and hence a fish assemblage that is relatively consistent across different structure types.

Artificial reefs, like other manmade structures such as oil and gas platforms (Love et al., 2019; Harvey et al., 2021a), pipelines (Love and York, 2005; Schramm et al., 2021), and wharf and jetty infrastructure (Scanlon et al., 2024) are all important fish habitats. However, they are challenging to sample due to their limited, and often small size and close spatial proximity. This limits the numbers of spatially independent samples that can be obtained but does not take away from the fact that we need to understand how different structures and/or different designs constructed from different materials affect fish assemblages. This was a limitation in our study, particularly when sampling Abitats and Pyramids, as only one spatially independent replicate could be collected at each module due to their small size, and the total number of each module was low. This limitation should be considered when designing artificial reefs, to allow statistically rigorous assessment of their influence on marine communities.

This study supports previous research (summarised in Pondella et al., 2022) by highlighting the importance of structural complexity in shaping the fish assemblages associated with ARs. Including features with high vertical relief and/or high structural complexity may be beneficial for ARs that aim to enhance local recreational fishing opportunities. Furthermore, this study highlighted the importance of species-specific preferences for different structural features, which can be exploited in AR designs to promote higher abundances of species that fulfil the AR's purpose (e.g., targeted species for the enhancement of recreational fishing). Importantly, the learnings and results of this study may be used to assist in optimising fish production by implementing environmentally benign designs in new structures such as offshore wind farms and ports to optimise the social, environmental, and economic outcomes (Harvey et al., 2021a; Elrick-Barr et al., 2022). This collection of findings should be considered in the planning phase of AR projects to ensure the resultant structures are more likely to meet their goals.

CRediT authorship contribution statement

Brooke T. Marshall: Writing – review & editing, Writing – original draft, Formal analysis, Conceptualization. **Sam R. Russell:** Writing – review & editing, Investigation. **James H. Florisson:** Writing – review & editing, Funding acquisition. **Benjamin J. Saunders:** Writing – review & editing, Formal analysis, Conceptualization. **Stephen J. Newman:** Writing – review & editing, Writing – original draft, Conceptualization. **Euan S. Harvey:** Writing – review & editing, Writing – original draft, Investigation, Funding acquisition, Formal analysis, Conceptualization.

Ethics

All data collection was performed according to the Australian Code of Practice for the care and use of animals for scientific purposes. This research was undertaken under the Curtin University Animal ethics approval ARE 2018-20.

Declaration of competing interest

SRR and JHF work for Recfishwest who received funding from BHP and Woodside.

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Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: James H. Florisson reports financial support was provided by Woodside Energy Ltd. James H. Florisson reports financial support was provided by BHP Group Ltd. Sam R. Russell reports financial support was provided by Woodside Energy Ltd. Sam R. Russell reports financial support was provided by BHP Group Ltd. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.marenvres.2025.107103.

Data availability

Data will be made available on request.

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